Supplemental Online Content

HIP Trial Investigators. Effect of early vs late inguinal hernia repair on serious adverse event rates in preterm infants: a randomized clinical trial. *JAMA*. Published March 26, 2024.
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This supplemental material has been provided by the authors to give readers additional information about their work.

eTable 1. Additional information related to trial enrollment

	No. of infants
Determined at site level to be ineligible due to "associated factor affecting the timing of hernia repair"	
Incarceration during NICU	10
Infant had multiple prior abdominal operations	5
Severe respiratory disease	5
Hernia contains ovary, preferred early	4
Severe cardiac disease	4
Airway anomaly requiring operation	2
Research team contacted to close to anticipated discharge, enrollment not feasible	2
Associated undescended testicle	2
Viral upper respiratory infection	2
COVID restriction (no elective operation allowed)	2
Transferring to a hospital closer to home that is not participating in the study	2
On anticoagulant therapy	1
Very large hernia; surgeon preferred early repair	1
Infant considered too small	1
Unsure if hydrocele or hernia, prefer to wait	1
Want to coordinate for ophthalmology procedure	1
Coordinating with neurosurgery (shunt placement)	1

Planning for G tube placement	1
Severe pulmonary hypertension	1
Unexplained thrombocytopenia	1
Await discontinuation of steroids	1
Believed hernia was causing feeding intolerance	1
Reasons for refusal of randomization by parents/guardians (n=613)	
Preferred early repair	280
Preferred late repair	196
Preferred that physician decided timing of hernia repair	71
No reason given	66
Reasons for refusal of randomization by physician (n=37)	
Preferred early repair	14
Preferred late repair	14
No reason given	9
“Other reasons” for eligible infants not being consented (n=16)	
Non-English speaking and consent forms only available in English	4
COVID restrictions	4
Custody issues	3

Hernia incarceration concerns while in NICU	2
Insurance reasons	1
Medically unstable	1
Transfer for cardiac surgery	1

eTable 2. Frequentist primary and major secondary outcome analyses

Outcome	Early Repair (n=159)	Late Repair (n=149)	Risk difference (95% confidence interval)	P value	Relative risk (95% confidence interval)	P value
Primary outcome						
Infant had ≥1 SAE	44 (28%)	27 (18%)	-9.0 % (-16.5, -2.0 %)	0.01	0.65 (0.46 – 0.92)	0.01
Secondary outcome						
Total hospital days during study period, median (IQR)	19.0 (9.8, 35)	16.0 (7, 38)			0.91 (0.74-1.12)	0.36

A logistic mixed-effect model was used to analyze the primary outcome, and a negative binomial mixed model for total hospital days. All models included gestational age group as a covariate and center as a random intercept. Frequentist and Bayesian analyses used the same models except for the frequentist analysis of the primary outcome which used a generalized estimating equation logistic model (exchangeable correlation for center) due to non-convergence of the mixed-effect model. The participating centers in the HIP trial were all large pediatric healthcare centers with pediatric surgery services and almost all have an existing University affiliation. All surgeons involved were pediatric general and thoracic surgeons and there were no pediatric urologists involved. The number of surgeons at each site ranged widely as expected. The number of infants randomized by each site is given below.

eTable 3. Enrollment by center

Center	No. (%)		
	Early Repair n = 172	**Late Repair** n = 166	**Overall** N = 338
Akron Children's Hospital	2 (1.2%)	2 (1.2%)	4 (1.2%)
Albany Medical Center	4 (2.3%)	2 (1.2%)	6 (1.8%)
All Children's Johns Hopkins	3 (1.7%)	4 (2.4%)	7 (2.1%)
Amplatz Children's Hospital	5 (2.9%)	5 (3.0%)	10 (3.0%)
Arkansas Children's Hospital	16 (9.3%)	17 (10%)	33 (9.8%)
Cardinal Glennon/SLU	7 (4.1%)	5 (3.0%)	12 (3.6%)
Children's Hospital of Alabama	3 (1.7%)	3 (1.8%)	6 (1.8%)
Children's Hospital of Wisconsin	2 (1.2%)	0 (0%)	2 (0.6%)
Children's Memorial Hermann Hospital	7 (4.1%)	8 (4.8%)	15 (4.4%)
Cincinnati Children's Hospital	2 (1.2%)	0 (0%)	2 (0.6%)
Cohen Children's Medical Center	0 (0%)	1 (0.6%)	1 (0.3%)
Columbia University Medical Center	5 (2.9%)	5 (3.0%)	10 (3.0%)
Connecticut Children's Medical Center	4 (2.3%)	4 (2.4%)	8 (2.4%)
Dartmouth-Hitchcock Medical Center	2 (1.2%)	0 (0%)	2 (0.6%)
Dayton Children's Hospital	5 (2.9%)	5 (3.0%)	10 (3.0%)
Duke Children's Hospital	11 (6.4%)	9 (5.4%)	20 (5.9%)
Erlanger/UT Chattanooga	11 (6.4%)	10 (6.0%)	21 (6.2%)
Le Bonheur Children's Hospital	8 (4.7%)	8 (4.8%)	16 (4.7%)
Medical University of South Carolina Children's Hospital	7 (4.1%)	7 (4.2%)	14 (4.1%)
Nationwide Children's Hospital	9 (5.2%)	8 (4.8%)	17 (5.0%)
Naval Medical Center San Diego	1 (0.6%)	2 (1.2%)	3 (0.9%)
Penn State Milton S. Hershey MC	0 (0%)	1 (0.6%)	1 (0.3%)
Rady Children's Hospital/UCSD	1 (0.6%)	1 (0.6%)	2 (0.6%)
Seattle Children's Hospital	1 (0.6%)	0 (0%)	1 (0.3%)
Shands Hospital for Children	1 (0.6%)	1 (0.6%)	2 (0.6%)
Southern California Permanente Group	6 (3.5%)	7 (4.2%)	13 (3.8%)
St. Louis Children's Hospital	2 (1.2%)	1 (0.6%)	3 (0.9%)
Texas Children's Hospital	3 (1.7%)	1 (0.6%)	4 (1.2%)
Tufts University	1 (0.6%)	2 (1.2%)	3 (0.9%)
UCLA	1 (0.6%)	2 (1.2%)	3 (0.9%)
UCLA-Harbor	1 (0.6%)	2 (1.2%)	3 (0.9%)
University of Iowa Children's Hospital	11 (6.4%)	13 (7.8%)	24 (7.1%)
University of Nebraska Medical Center	0 (0%)	1 (0.6%)	1 (0.3%)
University of Virginia	3 (1.7%)	4 (2.4%)	7 (2.1%)
Utah/Primary Children's	3 (1.7%)	3 (1.8%)	6 (1.8%)
Valley Children's	4 (2.3%)	2 (1.2%)	6 (1.8%)
Vanderbilt Children's Hospital	11 (6.4%)	13 (7.8%)	24 (7.1%)
Women & Children's Hospital of Buffalo	6 (3.5%)	6 (3.6%)	12 (3.6%)
Women & Infants Hospital Rhode Island	3 (1.7%)	1 (0.6%)	4 (1.2%)

eAcknowledgments

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